

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and deployed a **Python**-based anomaly detection model using **scikit-learn** on a dataset of **~150,000 transactions**, achieving an AUC of **0.82**.
- Engineered **Python** data pipelines on **AWS Lambda** and **S3** to process and transform data for machine learning workflows, handling approximately **2,000 events per second** during peak hours.
- Designed and implemented **Java Spring Boot** REST APIs for inter-service communication, incorporating request validation and circuit-breaker patterns to enhance system resilience.
- Collaborated with enterprise customers to gather requirements for **AI applied research** solutions, translating business needs into technical specifications for model development.
- Contributed to the development of novel **transformer neural networks** by implementing and testing experimental model architectures within a research framework.
- Supported the ecosystem by developing training materials and running advisory sessions on the application of **machine learning algorithms** for industry partners.
- Containerized microservices using **Docker** and managed deployments on a **Kubernetes** cluster, configuring basic resource limits for stable operation.

SKILLS

Machine Learning – ML algorithms, transformer neural networks, anomaly detection, classification, regression, model evaluation, scikit-learn, R, RStudio

Programming Languages – Python, Java, SQL, R

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, Git

Backend Frameworks – Spring Boot, FastAPI, Flask

Data Analysis & BI – pandas, NumPy, Power BI, Tableau, DAX, Power Query, data modelling, data warehousing

Databases – PostgreSQL, MySQL, Oracle

PROJECTS

AI Applied Research for Customer Impact Measurement

- Developed a predictive model using **Python** and **scikit-learn** to forecast customer engagement, trained on a dataset of **~120,000 user interaction** records and achieving an F1 score of **0.75**.
- Built data pipelines in **Python** to extract and aggregate key features from diverse data sources, supporting the evaluation of ML model performance against real-world business metrics.
- Presented findings and model performance metrics to research stakeholders, detailing the methodology and the measurable impact of the applied AI solutions.

Machine Learning Model Deployment Framework

- Containerized ML inference services using **Docker** and orchestrated deployments on an **AWS**-managed Kubernetes environment.
- Implemented CI/CD pipelines with **Jenkins** to automate the build, test, and deployment of ML models, reducing manual deployment time by **~40%**.
- Developed RESTful APIs using **FastAPI** to serve model predictions, ensuring low latency and high availability for downstream applications.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)